

Comparison Operators

In R we can use comparison operators to compare variables and return logical values. Let's see some relatively self-explanatory examples:

Greater Than

In [3]:

```
5 > 6
```

Out[3]:

```
FALSE
```

In [7]:

```
6 > 5
```

Out[7]:

```
TRUE
```

We can also do element by element comparisons for two vectors:

In [5]:

```
v1 <- c(1,2,3)
v2 <- c(10,20,30)
```

In [6]:

```
v1 < v2
```

Out[6]:

```
TRUE TRUE TRUE
```

Greater Than or Equal to

In [8]:

```
6 >= 6
```

Out[8]:

```
TRUE
```

In [10]:

```
6 >= 5
```

Out[10]:

```
TRUE
```

In [11]:

```
In [11]:
```

```
6 >= 7
```

```
Out[11]:
```

```
FALSE
```

Less Than and Less than or Equal To

```
In [12]:
```

```
3 < 2
```

```
Out[12]:
```

```
FALSE
```

```
In [13]:
```

```
2 <= 2
```

```
Out[13]:
```

```
TRUE
```

Be very careful with comparison operators and negative numbers! Use spacing to keep things clear. An example of a dangerous situation:

```
In [14]:
```

```
var <- 1
```

```
In [15]:
```

```
var
```

```
Out[15]:
```

```
1
```

```
In [16]:
```

```
# Comparing var less than negative 2  
var < -2
```

```
Out[16]:
```

```
FALSE
```

```
In [17]:
```

```
# Accidentally reassigning var!  
var <- 2
```

```
In [18]:
```

```
var
```

```
Out[18]:
```

Out[18]:

2

Keep syntax highlighting in mind to help you avoid this mistake!

Not Equal

In [19]:

```
5 != 2
```

Out[19]:

TRUE

In [26]:

```
5 != 5
```

Out[26]:

FALSE

Equal

In [27]:

```
5 == 5
```

Out[27]:

TRUE

In [28]:

```
2 == 3
```

Out[28]:

FALSE

Vector Comparisons

We can apply a comparison of a single number to an entire vector, for example:

In [29]:

```
v <- c(1,2,3,4,5)
```

In [30]:

```
v < 2
```

Out[30]:

TRUE FALSE FALSE FALSE FALSE

In [31]:

```
v == 3
```

Out[31]:

```
FALSE FALSE TRUE FALSE FALSE
```

Later on we will see how we can use these vector comparisons to select parts of a vector!